Baseline Reproduction - Technical Memo

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Subject: Baseline Reproduction - Predicting Academic Dropout

1. Repository Submission

- **Repository URL:** https://github.com/JasonB2004/csci3052u-ml-group-project.git
- **Release Tag:** v1.0-baseline

2. Reproducible Environment & Repository Setup

The project uses Python 3.13.15 and requirements.txt with the package versions. We used the random seed 42 (same as Milestone 2) to reproduce the main data split. We also save the split indices and results.

To run the project, you will need to install the required packages using **requirements.txt**, then run **python -m tests.test_pipeline** and **python -m src.run_baselines**. The second command will regenerate the baseline results.

3. Working Data Pipeline

The data pipeline is in src/data.py. It uses the same cleaning steps as in Milestone 2. It goes through 4,424 student records and predicts Dropout, Enrolled, or Graduate for each.

Cleaning: We rename and clean the columns, remove irrelevant information and second-semester features, and we prepare occupation categories.
Splitting: We use a stratified 70/15/15 training/validation/test split (3,096/664/664 students).
Preprocessing: We use One-Hot encoding to encode categorical features and normalize only the training data. The dataset has 169 features.

4. Baseline Methodology

We compared two benchmarks with a course-level classification model:

Majority Class: This always predicts Graduate, which is the most common outcome.
Stratified Random: This predicts random outcomes based on their proportions in the training data.
Course-Level Baseline: We used Softmax regression in NumPy and trained it using gradient descent. We tested different learning rates and regularization values with only the validation set.

5. Baseline Table

The results below are from the validation set. We used macro-F1 to give equal weight to all three outcomes.

Model	Accuracy	Macro-F1
Majority Class	0.500	0.222
Stratified Random	0.373	0.313
Softmax Regression	0.742	0.661

Softmax performed the best and exceeded our target from Milestone 2, which was 0.272 macro-F1. It averaged 0.667 ± 0.020 macro-F1. It was, however, less successful at identifying Enrolled students.

6. Discrepancy Analysis

We compared our results with the expected benchmarks and an independent scikit-learn model.

Match: The majority-class score matched the expected result at 0.222. The dataset size and split also matched Milestone 2.
Small Difference: Stratified random was close to the expected result of 0.333. It averaged 0.343 macro-F1 over 10 splits.
Discrepancy: Our NumPy model had a slightly different result from scikit-learn without regularization because they stopped training at different points.

7. Start / Stop / Continue Plan

● **Start:** We will start comparing k-nearest neighbours and decision trees in Milestone 4. We will also look for ways to improve our predictions for Enrolled students.

● **Stop:** We will stop using only accuracy to judge performance because it can hide poor results for individual classes.

● **Continue:** We will continue using the same pipeline, random seeds, validation-only model selection, and automated tests.

AI use: Claude Code helped with coding, tests, the notebook, and some documentation. Every team member reviewed the code before submission.